BERISSA MUYIZERE

CONTACTS

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LinkedIn

Github

EDUCATION

AkiraChix
codeHive - Diploma in Information Technology
February 2025 - Present

SKILLS

Programming Languages: Python, SQL
IoT communication protocol: MQTT
Sensor selection and interfacing.
Libraries and frameworks: Pandas, NumPy, Matplotlib, Seaborn, FastAPI, Django REST, MQTT, Framework.
IOT Tools: Tinker CAD, Onshape, HiveMQ.
Database management system Tools: DBeaver.

REFERENCES

Linda Kamau
Founder & Executive, AkiraChix
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Pascaline Kayitete Umulisa
Software Engineer, Ampersand E-Mobility
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Caroline Mutua,
Product Manager, Microsoft
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PERSONAL STATEMENT

Faced with the urgent reality of climate change, Berissa is inspired by how technology is opening new paths toward a sustainable future. Awareness alone isn't enough; she sees a powerful opportunity to turn technical knowledge into real-world solutions that reduce harm, restore balance, and create a future where both people and the planet can thrive.

PROJECTS

Carbon Track
Carbon Track is a real-time carbon emissions monitoring system for the Kenyan Tea Development Agency(KTDA) tea factories in Kenya. The system uses IoT sensors to send emissions data to a web portal, which provides tracking and reporting of CO2 emissions and supports regulatory compliance across the tea factory.

- Integrated ESP32 microcontrollers to interface with sensors and collect carbon emissions data for real-time monitoring and accurate measurement of the CO2 emissions.
- Implemented a real-time IoT data system that improved data handling for ESP32-based sensors.
- Used the MQTT protocol via HiveMQ broker for low-latency transmission of sensor data with a WebSocket server to enable continuous, real-time streaming of sensor data to the frontend, ensuring reliable data delivery speed for an improved user experience.
- Developed a system that collects and stores emissions data in PostgreSQL, and built a dashboard for data visualization with server-side rendering.
- Programmed a CO2 emissions monitoring device to detect emission threshold breaches and deliver alerts for compliance breaches.
- Designed a protective enclosure for sensor hardware using Onshape CAD software, improving durability and operational integrity in the field.

AgriTech
AgriTech is a system that facilitates access to and tracking of farm equipment for Rwandan cooperatives, improving equipment availability and operational efficiency by connecting cooperatives with the equipment suppliers and providing a tracking device for the farm equipment.

- Integrated ESP32 microcontrollers with a GPS module for real-time location tracking of farm equipment.
- Managed data transmission using the MQTT protocol via HiveMQ broker for real-time sensor data flow.
- Improved data reliability and lower latency by continuously sending the equipment location to the frontend before storing it, using a WebSocket server.
- Implemented Leaflet.js mapping library on a dashboard to display live equipment coordinates dynamically.

Titanic Survival Prediction – Data Cleaning and Classification Modeling
Developed a model to predict Titanic passenger survival using key demographic and travel features using the Titanic dataset, while analyzing patterns by gender, class, and cabin location for improved accuracy and application.

- Processed and cleaned data using beautifulsoup, handling missing values in age, cabin location, and embarked, using mean and mode to ensure completeness, and used skewness to reduce bias.
- Extracted titles, created family size and age groups, and encoded categorical variables using Scikit-learn preprocessing to transform raw data into meaningful features that highlight important social and demographic patterns, such as the number of spouses and children influencing survival chances.
- Visualized patterns with Matplotlib and Seaborn, identifying gender, class, and age as key predictors to ensure a balanced assessment.
- Built a logistic regression model in Scikit-learn to analyze relationships between key features and survival outcomes, improving prediction accuracy and enabling data-driven insights to identify factors affecting passenger survival during emergencies.
- Assessed model evaluation and tuning with Accuracy, F1-score, ROC-AUC, and optimized using GridSearchCV with cross-validation to ensure balanced performance and reduce overfitting, enabling reliable generalization to new data.
- Generated insights and Application on key survival factors (gender, class, age) and highlighted use cases for decision-making and risk analysis, supporting targeted emergency response and resource prioritization.